

ILLUSTRATION 3.45

Sheet 22

Sold Date		
Location	2024	2025
Boston		
Charlotte		
Chicago		
Dallas		
Miami		
Phoenix		
Salt Lake City		
Seattle		

☒ **Keep only**
☐ Exclude
Hide
☐ Drill down
Format...
Rotate Label
☒ Show Header
Show Missing Values

ILLUSTRATION 3.46

Sold Date	
Location	2025
Boston	-88,
Charlotte	-94,
Chicago	-58,
Dallas	-106,
Miami	-57,786
Phoenix	-20,787
Salt Lake City	-46,867
Seattle	-140,206

✓ Keep Only × Exclude     

8 items selected Difference in SUM(Gross Sales): **-614,679**

2025

ILLUSTRATION 3.47

Sold Date

Location	2025
Seattle	-140,206
Dallas	-106,783
Charlotte	-94,794
Boston	-88,837
Chicago	-58,619
Miami	-57,786
Salt Lake City	-46,867
Phoenix	-20,787

ILLUSTRATION 3.48

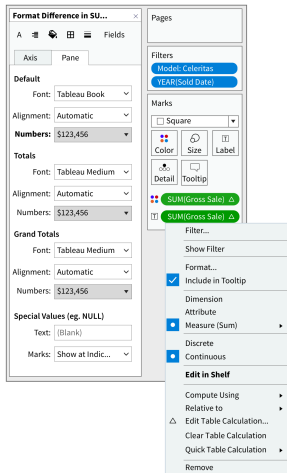


ILLUSTRATION 3.49

Sheet 22

Location	\$
Seattle	\$
Dallas	\$
Charlotte	\$
Boston	\$
Chicago	\$ (-58,619)
Miami	\$ (-57,786)

Edit Title...
Reset Title
Hide Title

Format Title...

ILLUSTRATION 3.50

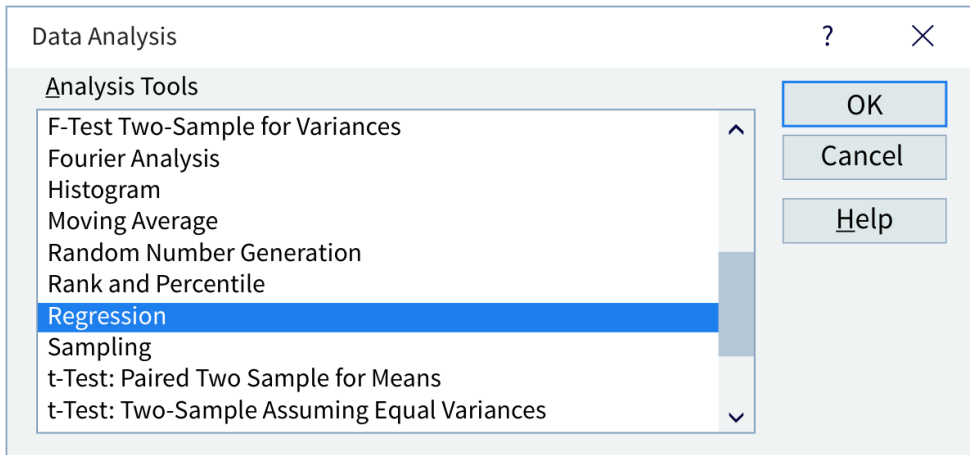


ILLUSTRATION 3.50_1

Sheet 22

Location	Sold Date 2025
Seattle	\$ (140,206)
Dallas	\$ (106,783)
Charlotte	\$ (94,794)
Boston	\$ (88,837)
Chicago	\$ (58,619)
Miami	\$ (57,786)
Salt Lake City	\$ (46,867)
Phoenix	\$ (20,787)

Edit Title

Tableau Light

15

B

I

U



Insert



Change in Gross Sales from 2024-Celeritas Model

Reset

OK

Cancel

Apply

ILLUSTRATION 3.51

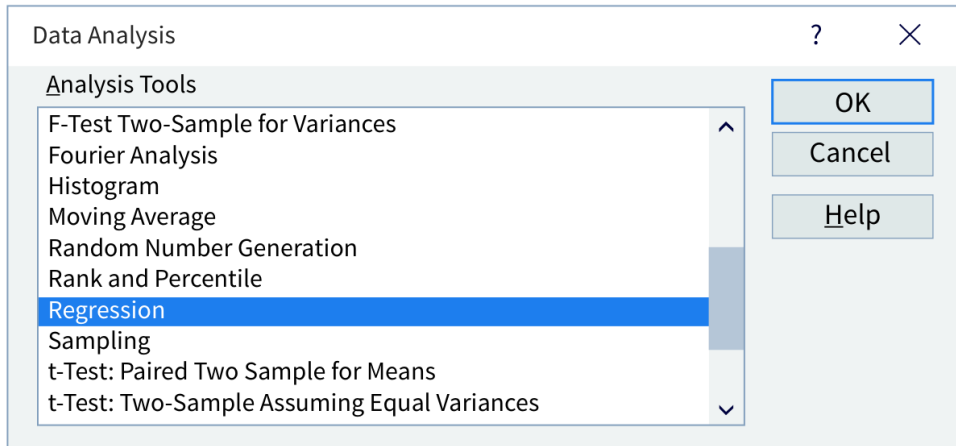


ILLUSTRATION 3.52

The screenshot shows an Excel spreadsheet with a data table and a Regression dialog box open. The data table has columns A, B, C, D, and E. Column A is labeled 'Month/Year' and column B is labeled 'Machine Hours'. The data spans from row 2 to row 22. The Regression dialog box is titled 'Regression' and has several sections: 'Input', 'Output options', 'Residuals', and 'Normal Probability'. The 'Input' section has 'Input Y Range' set to '\$D\$1:\$D\$37' and 'Input X Range' set to '\$B\$1:\$C\$37'. The 'Labels' checkbox is checked, and 'Constant is Zero' is unchecked. The 'Confidence Level' is set to 95%. The 'Output options' section has 'Output Range' set to '\$H\$3', 'New Worksheet Ply' is selected, and 'New Workbook' is unselected. The 'Residuals' section has 'Residuals', 'Standardized Residuals', 'Residuals Plots', and 'Line Fit Plots' all unchecked. The 'Normal Probability' section has 'Normal Probability Plots' unchecked. The dialog box has 'OK', 'Cancel', and 'Help' buttons.

	A	B	C	D	E
1	Month/Year	Machine Hours	Maintenance	Maintenance	
2	Jan-23	3250			
3	Feb-23	3770			
4	Mar-23	2470			
5	Apr-23	4030			
6	May-23	4940			
7	Jun-23	4290			
8	Jul-23	5330			
9	Aug-23	4550			
10	Sep-23	2600			
11	Oct-23	4810			
12	Nov-23	6110			
13	Dec-23	5460			
14	Jan-24	3270			
15	Feb-24	3790			
16	Mar-24	2490			
17	Apr-24	4050			
18	May-24	4960			
19	Jun-24	4310			
20	Jul-24	5350			
21	Aug-24	4570			
22	Sep-24	2620			

ILLUSTRATION 3.53

SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.978687872				
R Square	0.95782995				
Adjusted R Square	0.955274189				
Standard Error	1283.821342				
Observations	36				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>
Intercept	5252.858452	897.8427419	5.850533	1.5E-06	3426.183659
Machine Hours	3.569202976	0.43769194	8.154601	2.05E-09	2.678712028
Maintenance Request	759.8366754	168.7763028	4.502034	7.94E-05	416.4587054

Phần Bài tập Ngắn

Brief Exercises (BE)

Motivation	Event
1. Opportunity	
2. Regulation or law	
3. Problem	
4. Process or performance evaluation	

Motivation to Perform Analyses	Analytics Area
1. Analyze the mean, median, and mode number of failed login attempts after the company changed password requirements to understand if the benefits of enhanced password requirements outweigh the costs.	
2. Analyze the correlation between the quantity of phishing attempts reported and employee attendance at cybersecurity training to determine if training is related to reporting of phishing attempts.	
3. Analyze trends of failed login attempts over time to determine if employees are complying with company policy regarding login credentials.	
4. Perform a linear regression analysis to consider the likelihood of future information security breaches based on the independent variables of dollars spent on cybersecurity training and encryption software.	
5. Perform a what-if analysis to identify the appropriate dollar amount of spending that is necessary to meet the entity's goals for information security.	

BE 3.3 & BE 3.5

Scenario	Motivation Source
1. You are a tax accountant working at a large multi-national company. Executive management is trying to decide the best country in which to expand operations. You have been tasked with identifying independent variables and performing regression analysis to predict potential revenue from this expansion.	
2. Your company has several key performance indicators (KPIs) associated with its manufacturing process – especially quality control. You have been tasked with performing an analysis to identify whether quality control benchmarks are being accurately recorded in the information system.	
3. You are an external auditor working on a public company manufacturing client. You have been asked to test the operating effectiveness of internal controls associated with the purchasing groups' compliance with the internal control that all purchase orders above \$10,000 are approved by a purchasing supervisor.	
4. You are a financial analyst working for a company that distributes consumer products to retailers across the U.S. You must find out why sales of your historically most popular product, the electric grill, have been declining.	

FY	Activity	Department	Check Date	Vendor	Check Total
2024	Landfill Management	Sanitation Svcs	8/22/2024	Smith Temporaries, Inc	\$ 570.24
2025	Service Maintenance Centers	Public Works and Transportation	8/14/2025	Ferrell Gas, LP	\$ 32.22
2023	Water Treatment	Water Utilities	10/30/2023	Custom-Crete Redi-Mix, LLC	\$ 100.00
2023	Inventory	Inventory Purchase	11/21/2023	Installer Sales and Service	\$ 8.76
2024	Fleet Replacement	Equipment and Fleet Management	6/28/2024	Sam Pack's Five Star Ford	\$ 22,155.00
2024	Homeless Housing Services	Housing/Community Services	6/26/2024	Saint Augustine Estates	\$ 614.00
2024	Complete Street	Public Works and Transportation	3/11/2024	HERC Rentals Inc	\$ 4,236.00
2023	Non-Departmental	Office of Budget	12/24/2023	William H Bancroft Jr.	\$ 2,000.34

BE 3.6 & BE 3.7

Initial Descriptive Analysis Questions or Sub-Questions	Analysis Choice
1. How many units were produced for each product category in the facility?	
2. What is the average number of units produced by machine #1,065 during first shift, second shift, and third shift?	
3. What is the greatest number of units that machine #1,810 produced?	
4. What is the range of units produced during the period?	

Purchase Order No	Purchase Order Date	Vendor ID	Vendor Name	Item Code	Item Description	Item Quantity	Item Price	Total PO Amount	PO Approved?
2001	1/18/2025	13789	ABC Parts	V473	Cylinder Head	150	\$ 56.24	\$ 8,436.00	
2002	1/18/2025	13356	L&M Manufacturing	V302	Camshaft	250	\$ 57.56	\$ 14,390.00	Y
2003	1/18/2025	13780	TGIF Works	V292	Crankshaft	155	\$ 58.45	\$ 9,059.75	
2004	1/18/2025	13852	LP, LLP	K385	Oil Pan	153	\$ 36.24	\$ 5,544.72	
2005	1/18/2025	13513	GAPISH, CO	K395	Oil Valve	157	\$ 36.47	\$ 5,725.79	
2006	1/19/2025	13253	PTO Park, Co	M390	Nickel Metal Hydride Battery	201	\$ 18.46	\$ 3,710.46	
2007	1/19/2025	13850	STY Manufacturing	M392	Lithium Ion Battery	200	\$ 19.50	\$ 3,900.00	
2008	1/19/2025	13861	Bing Blitz, Inc.	Z389	Struts	268	\$ 37.89	\$ 10,154.52	Y

BE 3.8 & BE 3.9

Objective	Initial Question	Sub-questions	Possible Measures
Why did sales returns decrease from prior year?	1.	Are there anomalies in sales returns for any specific product the company sells? Are there identifiable trends in sales returns for any specific product?	Create a scatterplot comparing quantity returned to sales price for items to identify anomalies in returns. Create a scatterplot comparing sales dollars to sales returns for items to identify anomalies in returns. 2.
Why is there a large decrease in sales returns in the northwest region?	What factors are driving the decrease in sales returns in the northwest region?	3.	4.

Objective	Initial Question	Sub-questions	Possible Measures
Why did sales returns decrease from prior year?	1.	Are there anomalies in sales returns for any specific product the company sells? Are there identifiable trends in sales returns for any specific product?	Create a scatterplot comparing quantity returned to sales price for items to identify anomalies in returns. Create a scatterplot comparing sales dollars to sales returns for items to identify anomalies in returns. 2.
Why is there a large decrease in sales returns in the northwest region?	What factors are driving the decrease in sales returns in the northwest region?	3.	4.